

COS 135 Individual Assignment 4

Due: Monday 02/23/26 End of the day

What to submit:

- .c source codes for each task.

[10 pts] Comments are required in the following locations (in each C source code):

- At the top of each source code, comment your name and a short program description.
- Comment the purpose of each variable.
- Comment major sections of code such as input, processing, and output.

Program Design: Your program is a professional document and must be neat and easy to read. All programs should follow these specifications.

- Comments should be aligned and entered in a consistent fashion
- Blank lines should be added to aid readability
- Code within blocks should be indented
- Comments should not contain spelling mistakes
- Variables names should be meaningful

Write C programs for following tasks and submit your source codes (you must submit .c files without errors). Sample Program inputs are highlighted in yellow.

(a) [20 pts]

On the NASA Space Shuttle, three computers were in use simultaneously. Computations were performed by all three computers, and the result of each computation was put to a vote. There were two types of computations: critical and non-critical.

- If it was a critical computation (i.e., code **C**), all three computers must agree on the result (i.e., an integer). If the computers produced different results, the computation was not considered and discarded.
- If it is a non-critical computation (i.e., code **N**), at least two computers should provide the same result; that result was then used. But if all three computers each provided a different result, then the computation was discarded.

Write a program that prompts for three different inputs (a single character type-code and a whole number, a pair at a time) as a simulation of three different computational results. If the above conditions are satisfied, your program should output the selected value. If not, output the word “ERROR”. Your program should also output “ERROR” if a user inputs mixed types of computations.

<u>Sample output #1:</u> Enter first number: C 5 Enter second number: C 5 Enter third number: C 5 Value: 5	<u>Sample output #2:</u> Enter first number: C 5 Enter second number: C 5 Enter third number: C 9 Value: ERROR
<u>Sample output #3:</u> Enter first number: N 16 Enter second number: N 21 Enter third number: N 16 Value: 16	<u>Sample output #4:</u> Enter first number: C 9 Enter second number: N 9 Enter third number: C 9 Value: ERROR

* Note that there can be many other variations, sample outputs only show a few.

Example program execution:

```
nimesha@cos135:/media/cos135/a4_codes$ sudo gcc A4_a.c -o A4a
nimesha@cos135:/media/cos135/a4_codes$ ./A4a
Enter the first type number pair: C 2
Enter the second type number pair: C 2
Enter the third type number pair: C 2
Value: 2
nimesha@cos135:/media/cos135/a4_codes$ ./A4a
Enter the first type number pair: N 41
Enter the second type number pair: N 32
Enter the third type number pair: N 32
Value: 32
nimesha@cos135:/media/cos135/a4_codes$ ./A4a
Enter the first type number pair: N 23
Enter the second type number pair: N 21
Enter the third type number pair: N 20
ERROR
nimesha@cos135:/media/cos135/a4_codes$ ./A4a
Enter the first type number pair: N 11
Enter the second type number pair: C 11
Enter the third type number pair: N 11
ERROR
nimesha@cos135:/media/cos135/a4_codes$
```

(b) [20 pts]

Write a C program to request information from a user (via keyboard input) and outputs a summary statement.

In your C program, a user enters the following information of a new job applicant.

Program inputs:

New applicant's name:

New applicant's birth year:

New applicant's preferred monthly salary (USD):

Two examples (program outputs and interactive user inputs) are stated below:

Sample output # 1:

New applicant's name: Britney William
Enter applicant's birth year: 1982
Enter monthly salary (USD): 5000

Sample output #2:

New applicant's name: Curtly Ambrose
Enter applicant's birth year: 1976
Enter monthly salary (USD): 5500

Program outputs:

After obtaining the above information, the program should output following summary statements respectively:

Example 1 output:

Britney William is a 40 years old applicant and requests a monthly salary of \$5000.00

Example 2 output:

Curtly Ambrose is a 46 years old applicant and requests a monthly salary of \$5500.00

[5pts] Specific program requirements:

Define the following constant variable to calculate the age:

#define CURRENT_YEAR 2026

Example program execution:

```
nimesha@cos135:/media/cos135/a4_codes$ sudo gcc A4_b.c -o A4b
nimesha@cos135:/media/cos135/a4_codes$ ./A4b
New applicant's name: Tom Cruise
Enter applicant's birth year: 1972
Enter monthly salary (USD): 12000
Tom Cruise is a 51 year-old applicant, who requests a monthly salary of $12000.00.
nimesha@cos135:/media/cos135/a4_codes$ ./A4b
New applicant's name: Ian Botham
Enter applicant's birth year: 1955
Enter monthly salary (USD): 18800
Ian Botham is a 68 year-old applicant, who requests a monthly salary of $18800.00.
nimesha@cos135:/media/cos135/a4_codes$
```

(c) **[50 pts]**

The Nova Catering Company has asked you to write a program to produce a customer bill. The caterer provides 3 meals: beef meal for \$12.95, chicken meal for \$10.95, and vegan meal for \$8.95. The user is requested to enter the number of beef, chicken, and vegan meals. An 18% gratuity is added to the food cost.

Nova offers 4 banquet halls. Room 1 seats 200 and has a rental fee of \$250. Room 2 seats 150 and has a rental fee of \$200. Room 3 seats 100 and has a rental fee of \$100. Room 4 seats 30 and has a rental fee of \$50. A 6.5% state tax is added to the room cost.

The program should use **constant variables** for *gratuity* and *state tax* (hardcoded in your code). (Note: it's a good practice to use all UPPERCASE letters to declare a constant variable, e.g. `PI = 3.14`).

Input: The number of beef meals, chicken meals, and vegan meals should be read from the keyboard using clear prompts. If any of the numbers are negative, display an appropriate error message about invalid data and end the program. If the total number of meals exceeds 200, display an error message that there is no room large enough and end the program.

Processing:

Determine the **smallest size room that will accommodate the party** and compute room cost and tax. Compute the cost of each type of meal (if ordered), food gratuity, and total bill. If a meal type is not ordered do not include it in the output.

Output:

Your name should be the first line of output (*replace Student name with your name*). Produce an attractive bill similar to the sample shown that lists the total number in the party, cost of the room, room tax, number of beef meals and their total cost, number of chicken meals and their total cost, number of vegan meals and their total, gratuity, and the grand total. **Field widths** should be used to produce neat columns versus blanks or tabs. The **precision** should be set to show two decimal places.

You will turn in your source code and **two** screen shots using following inputs.

1. Sample run of a party with 50 beef, 25 chicken, and 35 vegan meals.
2. Sample run of a party with 0 beef, 0 chicken, and 25 vegan meals.

[5pts] Specific program requirements:

Define the following constant variables to calculate the age:

```
#define BEEF_PRICE 12.95
#define CHICKEN_PRICE 10.95
#define VEGAN_PRICE 8.95
#define GRATUITY_RATE 0.18
#define ROOM_TAX_RATE 0.065
```

* See next page for sample outputs

Sample Output #1:

```
nimesha@cos135:/media/cos135/a4_codes$ sudo gcc A4_c.c -o A4c
nimesha@cos135:/media/cos135/a4_codes$ ./A4c
Enter number of beef meals: 50
Enter number of chicken meals: 25
Enter number of vegan meals: 35

Nova Caterers by Student Name

Number in party                110
Room Cost $                    200.00
Room Tax $                     13.00
Number of Beef Meals           50
Cost of Beef Meals $           647.50
Number of Chicken Meals        25
Cost of Chicken Meals $        273.75
Number of Vegan Meals          35
Cost of Vegan Meals $          313.25
Gratuity $                     222.21
-----
Total $                        1669.71
nimesha@cos135:/media/cos135/a4_codes$
```

Sample Output #2:

```
nimesha@cos135:/media/cos135/a4_codes$ ./A4c
Enter number of beef meals: 0
Enter number of chicken meals: 0
Enter number of vegan meals: 25

Nova Caterers by Student Name

Number in party                25
Room Cost $                    50.00
Room Tax $                     3.25
Number of Vegan Meals          25
Cost of Vegan Meals $          223.75
Gratuity $                     40.28
-----
Total $                        317.27
nimesha@cos135:/media/cos135/a4_codes$
```